MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)

MG5DSCECT300/ MG5DSCECC300 – Digital Design using Verilog

Duration: 2 hour **Maximum Marks: 35**

Practical Examination – Model Question Paper

Instructions:

- Demonstrate the given task
- Viva session (Minimum 5 questions from the Practical Module)
- Lab Report

Evaluation Criteria:

- Viva: **10 marks**
- Lab Report: **5 marks**
- Design & Demonstration: **20 marks**

Tasks: (*Choose any one*)

1. Design & Simulate a **dataflow** model of a **2-input AND, OR, NOT, NAND, NOR, XOR and XNOR** gate using the **assign** statement; each with a **rise delay of 3ns and fall delay of 2ns**

2. Design & Simulate the basic gates (AND, OR, NOT) using any one of the universal **gates** in dataflow modeling.

3. Design & Simulate a **dataflow** model of a **Half-Adder** using the **assign** statement with a **rise delay of 2ns and fall delay of 3ns for Sum and Carry outputs separately**.

4. Design & Simulate a **dataflow** model of a **Full-Adder** using an **always block** (procedural**, using case/if-else**) without any delay.

5. Design & Simulate a **dataflow** model of a **Half-Subtractor** (Difference and Borrow)

6. Design & Simulate a **dataflow** model of a **4-to-2 priority encoder** using **conditional (?:) operators**

7. Design & Simulate a **dataflow** model of a **2-to-4 decoder** (with enable) using a **case statement**

8. Design & Simulate a **behavioral** model of a **4:1 Multiplexer using a case statement** inside an always block

9. Design & Simulate a **dataflow** model to convert a **4-bit Binary number to Gray code** using **XOR operations.**

10. Design & Simulate a **behavioral** model of an **SR flip-flop (with clock**) using an **always @ (posedge clk) block**.

11. Design & Simulate a **behavioral** model of a **T flip-flop derived from a D flip-flop** (T = J = K logic).

12. Design & Simulate a **dataflow** model of an **odd/even parity generator** for an **8-bit data word**

13. Design & Simulate a **behavioral** model of a **LIFO (Stack) register** of depth 8 using a **reg array**, with **push/pop operations** controlled by clock.

14. Design & Simulate a **behavioral** model of a **4-bit Up-Down counter** using an **always @ (posedge clk)** block with an **up/down control signal.**

15. Design & Simulate a **BCD (decade) up-counter** with a **ripple-carry-out (RCO) signal that goes high one clock before rollover.**

16. Design & Simulate a **dataflow** model of an **8-bit ALU** supporting **ADD, SUB, AND, OR, XOR** operations selected by a **3-bit opcode**, using the **conditional operator.**